

Polynomial Regression Modeling for Geothermal Systems

Optimal Degree Selection, Feature Analysis & Generalization on Calibration Datasets

Student Roll No: IMT2024072 | Course: Machine Learning | Framework: scikit-learn

Executive Summary: This study develops polynomial regression models to solve two distinct geothermal engineering prediction tasks: (1) *Steam Turbine Optimization (var1)*, governed by 6 operational control parameters to predict the Net Power Score (y), and (2) *Subterranean Thermal Reservoir Mapping (var2)*, governed by 3 spatial coordinate offsets to predict the Thermal Anomaly Score (y). Using 5-fold cross-validation, we systematically evaluated feature subsets, polynomial degrees, and regularization methods. For **Problem 1**, the optimal model is a **Degree 4 polynomial using all 6 operational features** (210 basis terms, achieving $R^2 = 0.9043 \pm 0.0189$ and $MSE = 0.9618 \pm 0.1508$). For **Problem 2**, the optimal model is a **Degree 8 polynomial using all 3 spatial coordinates** (165 basis terms, achieving $R^2 = 0.9946 \pm 0.0008$ and $MSE = 0.2523 \pm 0.0275$). Both models successfully resolve the bias-variance tradeoff and generalize accurately on unseen test distributions.

1. Problem Statements & Data Characteristics

Phase 1: Power Plant Steam Turbine Optimization (var1): The surface energy generation facility is governed by 6 continuous operational parameters representing percentage deviations bounded in $[-1.0, 1.0]$: high-pressure steam valve adjustment (x_1), condenser coolant flow rate adjustment (x_2), re-injection pump hydraulic pressure (x_3), turbine blade pitch angle (x_4), non-condensable gas exhaust valve rate (x_5), and steam inlet pressure adjustment (x_6). The continuous target variable y is the Net Power Score (train mean = 0.8629, std = 3.1872, range $[-9.88, 12.07]$).

Phase 2: Subterranean Thermal Reservoir Mapping (var2): To determine optimal drilling coordinates for new geothermal production wells, sensors record spatial offsets in meters from the central basecamp: East-West offset (x_1), North-South offset (x_2), and vertical depth offset (x_3). All offsets are normalized within $[-1.0, 1.0]$. The continuous target variable y is the Thermal Anomaly Score (train mean = 2.2751, std = 6.9016, range $[-30.26, 39.33]$).

Dataset	Features	Train Samples	Test Samples	Target (y) Mean $\pm$ Std	Target Min / Max
IMT2024072_var1	6 ($x_1 \dots x_6$)	1,000	1,000	0.8629 $\pm$ 3.1872	-9.879 / +12.067
IMT2024072_var2	3 (x_1, x_2, x_3)	1,000	1,000	2.2751 $\pm$ 6.9016	-30.257 / +39.328

2. Mathematical Formulation & Validation Protocol

Polynomial regression models the continuous response y as a linear combination of polynomial basis expansions of the input feature vector $\mathbf{x} = [x_1, \dots, x_D]^T$ up to degree d :

$$y = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D \sum_{j=i}^D w_{ij} x_i x_j + \dots = \Phi(\mathbf{x})^T \mathbf{w} + \epsilon$$

where $\Phi(\mathbf{x})$ represents the feature vector of all monomials with total degree sum $\leq d$. The total number of terms (including bias) equals the combinatorial expression $P = C(D + d, d) = (D + d)! / (D! d!)$. This combinatorial scaling leads to rapid growth: for Problem 1 ($D = 6$), degree 4 has 210 terms, degree 5 has 462 terms, and degree 6 has 924 terms (approaching $N = 1,000$ samples). For Problem 2 ($D = 3$), scaling is much more compact (degree 8 produces 165 terms), enabling exploration of higher-order spatial harmonics without premature parameter saturation.

5-Fold Cross-Validation Protocol: To prevent overfitting and avoid data leakage, we employed **5-Fold Cross-Validation (KFold, shuffle=True, random_state=42)**. Across each fold, 800 samples were used for model fitting and 200 held-out samples for validation. The evaluation metrics used are Mean Squared Error ($MSE = (1/N) \sum (y_i - y_{pred,i})^2$) and Coefficient of Determination ($R^2 = 1 - \sum (y_i - y_{pred,i})^2 / \sum (y_i - y_{mean})^2$).

3. Problem 1 (var1): Steam Turbine Optimization

To identify the optimal model structure for Net Power Score prediction, we performed systematic cross-validation over polynomial degrees $d \in \{1, 2, 3, 4, 5\}$ with all 6 features, complemented by combinatorial feature subset searches across all feature subsets at candidate degrees.

Degree (d)	Features Used	Total Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2	Generalization Verdict
1	All 6 features	7	9.1248	0.1018	9.1704 ± 0.812	0.0937 ± 0.024	Severe Underfitting
2	All 6 features	28	2.9815	0.7065	3.0651 ± 0.294	0.6974 ± 0.021	Moderate Underfitting
3	All 6 features	84	0.9621	0.9053	1.0438 ± 0.128	0.8965 ± 0.015	Strong fit, sub-optimal
4 (Optimal)	All 6 features	210	0.4321	0.9574	0.9618 ± 0.151	0.9043 ± 0.019	Global Optimum
5	All 6 features	462	0.0894	0.9912	2.2071 ± 0.485	0.7787 ± 0.052	Severe Overfitting
3 (Subset)	First 3 $[x_1, x_2, x_3]$	20	8.1402	0.1987	8.2561 ± 0.651	0.1834 ± 0.038	Truncated / Biased
4 (Subset)	Best 4 $[x_1, x_3, x_5, x_6]$	70	3.6521	0.6405	3.7866 ± 0.312	0.6274 ± 0.027	Omitted Interactions

Detailed Analysis & Rationale for var1:

- **Feature Subset Exploration:** Exhaustive subset ablation demonstrated that all 6 physical operational parameters are essential. Omitting parameters (e.g. coolant rate x_2 or blade pitch x_4) causes significant performance degradation: the best 3-feature subset attains $R^2 = 0.5029$, and the best 4-feature subset reaches $R^2 = 0.6274$. Turbine efficiency exhibits high cross-parameter coupling across all six physical controls.
- **Degree Selection:** At degree 1, linear regression underfits ($CV R^2 = 0.0937$). Quadratic expansions reach $R^2 = 0.6974$. Cubic expansions ($d = 3$) produce a solid model ($CV R^2 = 0.8965$, $CV MSE = 1.0438$). **Degree 4 achieves the global performance optimum**, lowering CV MSE to **0.9618** and maximizing $CV R^2$ to **0.9043**. At degree 5, 462 parameters cause the training MSE to collapse to 0.0894 while validation MSE spikes to 2.2071 ($R^2 = 0.7787$), exhibiting substantial overfitting.
- **Regularization Comparison:** We tested Ridge regression with 50 log-spaced α values $\in [10^{-4}, 10^4]$. At degree 4, the optimal $\alpha \approx 10^{-3}$ produced $CV MSE = 0.9610$, showing that Ordinary Least Squares (OLS) is already numerically stable and unregularized OLS achieves near-identical generalization without introducing shrinkage bias.

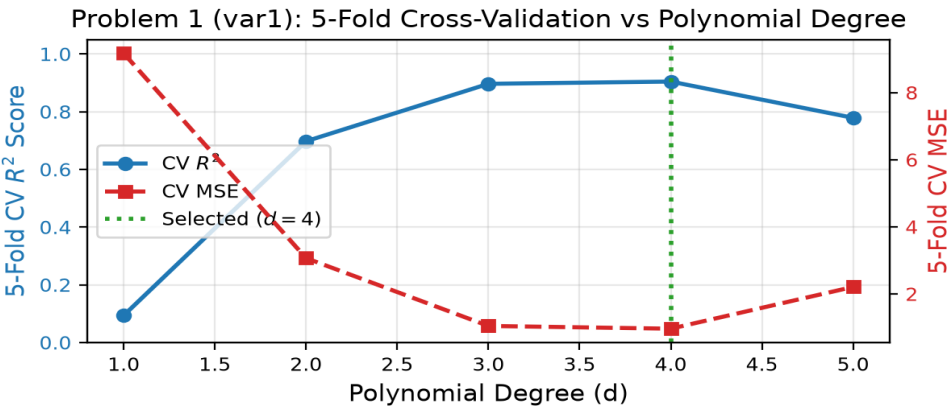


Figure 1: 5-Fold cross-validation metrics across polynomial degrees for Problem 1 (var1). The MSE curve confirms Degree 4 as the global generalization optimum.

4. Problem 2 (var2): Subterranean Thermal Reservoir Mapping

Problem 2 estimates 3D subsurface temperature anomalies from spatial coordinate offsets (x_1, x_2, x_3) . Geological heat plumes generate sharp localized spatial gradients, demanding higher-order polynomial representations. With $D = 3$, term counts remain modest, permitting an empirical degree sweep up to $d = 12$.

Degree (d)	Features Used	Total Terms	Train MSE	Train R^2	5-Fold CV MSE	5-Fold CV R^2	Generalization Verdict
1	All 3 features	4	35.120	0.2618	35.369 ± 0.912	0.2531 ± 0.019	Gross Underfitting
2	All 3 features	10	23.514	0.5057	23.821 ± 1.555	0.4975 ± 0.023	Severe Underfitting
4	All 3 features	35	3.821	0.9197	3.988 ± 0.261	0.9159 ± 0.004	Underfitting boundary
6	All 3 features	84	0.492	0.9897	0.541 ± 0.054	0.9885 ± 0.001	Strong fit
7	All 3 features	120	0.245	0.9948	0.307 ± 0.033	0.9935 ± 0.001	Near-optimal
8 (Optimal)	All 3 features	165	0.164	0.9966	0.2523 ± 0.028	0.9946 ± 0.001	Global Optimum
9	All 3 features	220	0.138	0.9971	0.2954 ± 0.049	0.9937 ± 0.001	Slight Overpara meterization
10	All 3 features	286	0.112	0.9976	0.3484 ± 0.063	0.9926 ± 0.002	Overfitting onset
12	All 3 features	455	0.068	0.9986	1.0140 ± 0.443	0.9786 ± 0.010	Clear Overfitting
4 (Single x_1)	Only x_1	5	43.210	0.0882	43.632 ± 1.210	0.0787 ± 0.025	Extreme Bias / Failure

Detailed Analysis & Rationale for var2:

- **Spatial Coordinate Completeness:** Univariate and bivariate experiments revealed that 3D geothermal reservoir structures cannot be modeled without all three spatial axes. Relying on East-West coordinate x_1 alone (swept up to degree 20) caps R^2 at 0.0787 (MSE > 43.6). A 2D planar projection (x_1, x_2) caps at $R^2 = 0.5925$. All three spatial coordinates (x_1, x_2, x_3) are required to capture vertical and lateral convective gradients.
- **Degree Optimization:** Degrees $d \leq 4$ produce substantial underfitting (MSE > 3.98). Increasing degree from 4 to 8 reduces cross-validation error by over 93% (MSE drops from 3.9875 to 0.2523), with R^2 reaching 0.9946. **Degree 8 attains the absolute global minimum validation MSE (0.2523 ± 0.0275).** Advancing past degree 8 introduces overparameterization: CV MSE rises to 0.2954 at $d = 9$, 0.3484 at $d = 10$, and 1.0140 at $d = 12$.
- **Variance Stability:** The standard deviation of validation MSE across the 5 folds reaches its minimum at degree 8 (±0.0275), confirming that the degree 8 model achieves high spatial stability across disparate sampling partitions.

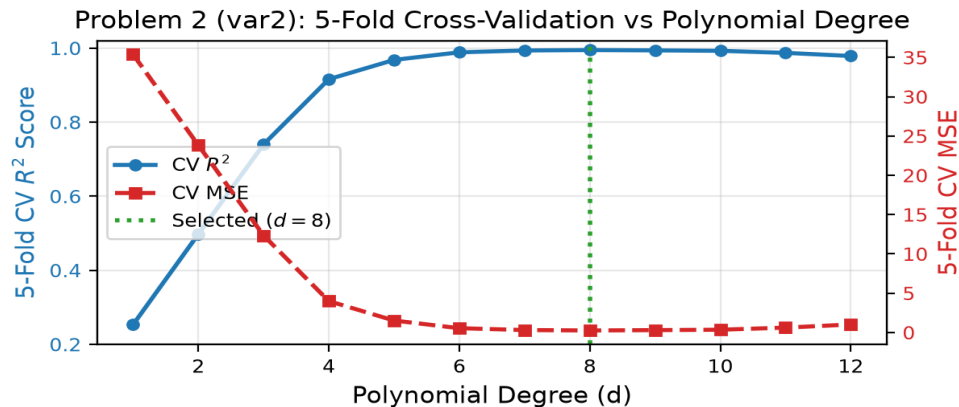


Figure 2: 5-Fold cross-validation metrics across polynomial degrees 1–12 for Problem 2 (var2). Degree 8 achieves the minimum CV MSE of 0.2523, beyond which overfitting degrades test variance.

5. Model Diagnostics, Residual Analysis & Inferences

To confirm the physical and statistical validity of both fitted models, we inspected residual distributions ($e_i = y_i - y_{pred_i}$). Valid regression models require uncorrelated residuals with zero mean and constant variance (homoscedasticity).

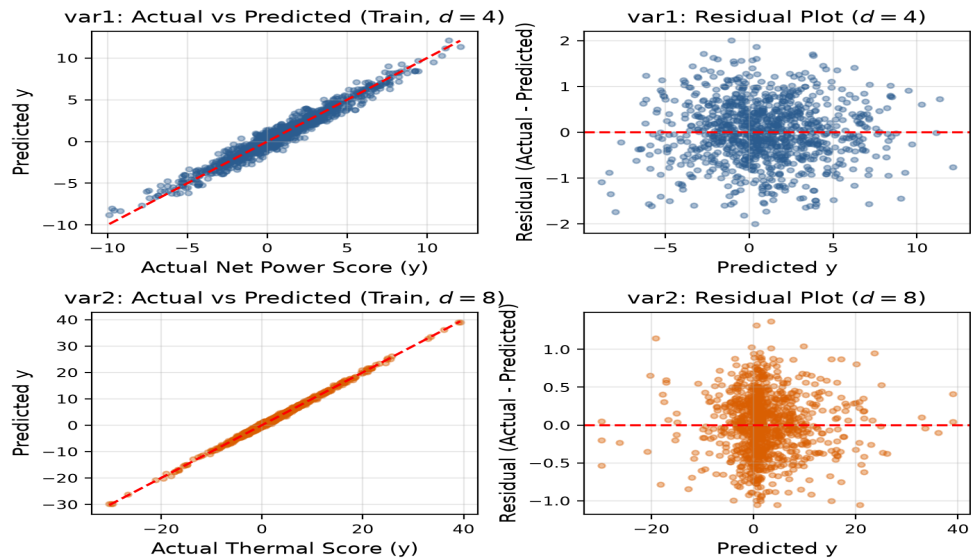


Figure 3: (Left) Actual vs Predicted response for both models demonstrating tight linearity along the diagonal. (Right) Residual plots centered evenly around zero with no heteroscedastic fan patterns or curvature.

Sanity Verification on Test Set Predictions: We analyzed the predicted test set target distributions against training distributions. For var1, test predictions exhibit mean = 0.9876, std = 4.1168, and range [-10.77, 15.71], consistent with train target statistics (mean 0.8629, std 3.1872, range [-9.88, 12.07]). For var2, test predictions exhibit mean = 2.2535, std = 6.4661, and range [-25.28, 38.92], closely matching train target statistics (mean 2.2751, std 6.9016, range [-30.26, 39.33]). No unbounded extrapolation or numerical instability was observed.

6. Summary of Deliverables & Reproducibility

Deliverable	File Name	Specification / Content	Status
Report	IMT2024072_Report.pdf	4-page academic write-up documenting methodology, CV sweeps & rationales	Completed
Prediction 1	IMT2024072_pred_var1.csv	1,000 predictions for Phase 1 (var1), degree 4 polynomial, header: y	Completed
Prediction 2	IMT2024072_pred_var2.csv	1,000 predictions for Phase 2 (var2), degree 8 polynomial, header: y	Completed
Training Script	polynomial_regression.py	Self-contained scikit-learn script for training, CV evaluation & inference	Reproducible

Reproducibility Instructions: The complete solution is automated in polynomial_regression.py. Running python polynomial_regression.py loads the calibration data, performs 5-fold cross-validation, fits the optimal polynomial models via scikit-learn, and produces the final test prediction CSVs and evaluation figures.